

Project Implementation Plan

Mistral 22B Fine-Tuning Pipeline for Audit Workpaper Intelligence

Document Type	Project Implementation & Onboarding Guide
Model	Mistral 22B (v24.09) — QLoRA Fine-Tuned
Domain	Finance, Audit Workpapers, Taxation & Compliance
Timeline	10 Weeks (+ Ongoing Feedback Cycles)
Prepared For	Product Manager / Leadership / Engineering Team
Date	April 2026
Classification	Internal Use Only — Confidential

How to Read This Document

This document is written for everyone — whether you are a **senior partner, an auditor, a software engineer, or a project manager**. No technical background is required to understand the flow of this project. Each phase is explained in plain language with clear steps, owners, and outcomes.

Non-Technical Reader	Read Phase descriptions and the 'What This Means' boxes. Skip the technical detail rows if needed.
Technical Reader	All step tables, configuration details, and infrastructure notes are written for you.
Project Manager	Focus on phase timelines, milestones, owners, and the risk register at the end.

Executive Summary

This project builds an **AI-powered audit workpaper assistant** for the firm. The system will read audit Excel workpapers, understand their content, and provide structured analysis — flagging risks, classifying findings, summarizing documents, and checking compliance thresholds — the same way a senior auditor would, but faster and at scale.

The AI model used is **Mistral 22B**, a state-of-the-art language model already deployed by major financial institutions including HSBC for compliance and risk automation. It will be fine-tuned (further trained) on the firm's own audit workpaper data to make it deeply familiar with the firm's specific standards, formats, and judgment patterns.

Critically, the entire system runs **on-premise** — on the firm's own servers. No client data ever leaves the building. No cloud service is involved. The firm owns the model, the data, and the output.

10 Weeks	~\$230	15M+	100%
Total project timeline	One-time fine-tuning cost	Training data records	On-premise — zero cloud

Project Overview — What Are We Building?

The Problem Being Solved

Audit workpapers today are reviewed manually. An auditor opens an Excel file, reads through trial balances, bank reconciliations, lead schedules, and findings — and applies professional judgment to each. This is time-consuming, prone to inconsistency across team members, and difficult to scale during busy season.

The Solution

We are building an AI assistant that sits alongside auditors. The auditor uploads or opens an Excel workpaper. The system converts it to a readable format, sends it to the AI model, and returns a structured analysis within seconds. The auditor reviews the AI output, makes corrections if needed, and those corrections feed back into improving the model over time.

What the System Will Do

Task	What the AI Does
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Risk Classification	Reads a workpaper and classifies each finding as Low, Medium, High, or Critical risk
Compliance Threshold Check	Checks if federal expenditure amounts cross thresholds (e.g. \$750K under 2 CFR 200.518) and flags them automatically
Finding Documentation	Drafts structured audit findings in the firm's standard format including condition, criteria, cause, effect, and recommendation
Workpaper Summarization	Produces a plain-language executive summary of any workpaper for partner review
Entity Recognition	Identifies and tags key entities — monetary values, regulation references, client names, dates, account codes

PHASE 1	Data Collection & Governance	Weeks 1 – 2
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What Happens in This Phase

Before any AI training begins, we collect and prepare the raw material — the firm's audit workpapers. This phase is about gathering the right data, ensuring it is legally cleared for use, scrubbing any sensitive client information, and setting up the security environment the entire project will run in.

Why It Matters	The quality of AI output is directly proportional to the quality of training data. Poorly prepared data produces a poorly performing model. This phase is the foundation of everything that follows.
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Step	Action	Detail	Owner
1.1	Identify Workpaper Sources	Collect Excel workpaper files across all major categories — SEFA, lead schedules, bank recs, trial balances, risk assessments, findings, internal controls, substantive testing. Target 100+ workpaper types.	Audit Team
1.2	Legal Sign-Off on Data Use	Obtain written approval from firm legal counsel confirming the workpapers (anonymized) can be used for internal AI training. This is a compliance requirement before any data is touched.	Legal / Partner

1.3	PII Scrubbing & Anonymization	Run automated anonymization pipeline. Remove all client names, taxpayer IDs, account numbers, addresses. Replace with placeholder tokens (e.g. CLIENT_001, ACCT_XXXX). Verify manually on a sample.	ML Engineer
1.4	Excel to JSON Conversion	Convert all .xlsx workpaper files to structured JSON format that the AI model can read. Each sheet becomes a structured object with row data, column headers, and sheet metadata preserved.	ML Engineer
1.5	VLAN & Security Setup	Set up air-gapped network segment for the training and inference servers. Configure firewall rules to block all external API calls. Enable TLS 1.3 for internal traffic only.	IT / DevOps
1.6	Data Quality Review	Senior auditor reviews a sample of 50-100 converted JSON workpapers to confirm the conversion preserved meaning and structure correctly. Fix any conversion issues found.	Senior Auditor

Excel to JSON — Plain English	An Excel workpaper with rows and columns gets converted into a text format the AI can read. Think of it like translating from a spreadsheet language into a reading language. The numbers, labels, and structure are all preserved — just in a different format.
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PHASE 2	Dataset Construction	Weeks 3 – 4
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What Happens in This Phase

This phase creates the actual training examples the model will learn from. Think of it as writing a textbook for the AI — showing it thousands of examples of what good audit analysis looks like, structured as question-and-answer pairs.

What is a Training Pair?	A training pair is one example: an INPUT (a workpaper excerpt or question) paired with the correct OUTPUT (what a senior auditor would say about it). The model studies thousands of these pairs to learn how to respond correctly.
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Step	Action	Detail	Owner
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2.1	Define Task Categories	Establish the 4 core tasks the model must perform: (1) Threshold & Compliance checking, (2) Finding Documentation, (3) Workpaper Summarization, (4) Risk Classification. Each task gets its own training examples.	ML Engineer + Senior Auditor
2.2	Author Training Pairs from 15M Dataset	Using the 15 million record dataset, extract and format question-answer pairs in Mistral's instruction format: [INST] question [/INST] answer. Target minimum 5,000 high-quality pairs across all 4 task types.	ML Engineer
2.3	Multi-Task Taxonomy Tagging	Tag each training pair with its task type, workpaper category, risk level, and regulation reference. This metadata enables targeted fine-tuning and later evaluation filtering.	ML Engineer
2.4	Senior Auditor Review of Training Examples	Senior auditors review and approve a minimum of 500 training pairs — especially for risk classification and findings documentation. Their corrections at this stage directly improve model quality.	Senior Auditor
2.5	Train / Validation / Test Split	Divide the full dataset: 80% for training, 10% for validation during training, 10% held back for final evaluation. The held-back 10% the model never sees until we test it.	ML Engineer
2.6	Format as JSONL	Save final dataset as JSONL (one training example per line). This is the file format the training software reads. Validate file integrity and record final pair counts per task.	ML Engineer

Training Pair Example — What One Looks Like

INPUT	CFDA 14.239, federal expenditures \$2.4M. Does major program testing apply?
OUTPUT	Yes — \$2.4M exceeds \$750K threshold under 2 CFR 200.518. Type A program requires major program risk assessment per AU-C 935.

PHASE 3	Infrastructure & LoRA Configuration	Weeks 5 – 6
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What Happens in This Phase

We set up the technical environment where training and serving will run. This includes configuring the two GPU servers, installing all required software, and setting the fine-tuning parameters that control how the model learns.

Plain English — What is Fine-Tuning?	The base Mistral 22B model already knows how to read and reason in English. Fine-tuning is like sending it to audit school — showing it thousands of real audit examples so it learns your firm's specific language, standards, and judgment patterns without forgetting everything it already knows.
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Server Configuration

	Server 1 — Training Server	Server 2 — Inference Server (Always On)
Purpose	Runs the fine-tuning job. Teaches the AI model your workpapers.	Serves the trained model to auditors in real time.
GPU Memory	48 GB VRAM — sufficient for Mistral 22B QLoRA training	48 GB VRAM — 22B model at 4-bit needs ~14 GB, giving ample headroom
Training Time	~18-20 hours per full fine-tuning run	Response time under 2 seconds per query
Software	Ubuntu 22.04, CUDA 12.2, HuggingFace PEFT, SFTTrainer	Ubuntu 22.04, Ollama + vLLM, REST API endpoint
Data Out?	No — fully on-premise, air-gapped VLAN	No — internal API only, no external calls ever

LoRA Fine-Tuning Parameters

What is LoRA?	Instead of retraining the entire 22B parameter model (extremely expensive), LoRA adds two tiny trainable layers into specific parts of the model. Only these tiny layers are updated during training. The result is a ~50MB adapter file that makes the model audit-aware, while the 14GB base model stays untouched.
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Parameter	Value	What It Means (Plain English)
Rank (r)	16	How many new audit-specific patterns the model can learn. 16 is the sweet spot for complex multi-task workpaper analysis.
Alpha (α)	32	Scaling factor — controls how strongly the new finance knowledge overrides the base model's general knowledge.
Target Modules	q_proj, v_proj	The specific attention layers inside the model where finance knowledge gets injected. Q = what to look for, V = what information to carry forward.

Quantization	4-bit NF4	Reduces model memory footprint from ~44GB to ~14GB. Makes it fit on a single server GPU without any quality loss.
Training Epochs	3	The model sees the entire training dataset 3 times. Enough to learn deeply without over-memorizing examples.
Adapter File Size	~50 MB	The output of fine-tuning — a tiny file that plugs into the base model and gives it all the audit knowledge. Easily versioned and updated.

PHASE 4	Training Run & Security QA	Weeks 7 – 8
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What Happens in This Phase

The actual AI training happens here. The model studies all the workpaper training examples, updates its internal knowledge, and produces the audit-aware adapter file. Simultaneously, the security team runs penetration testing to ensure no client data can leak through the model interface.

Step	Action	Detail	Owner
4.1	Baseline Benchmark	Before training begins, test the unmodified Mistral 22B on 50 sample workpaper questions. Record baseline scores. This is the 'before' measurement.	ML Engineer
4.2	Launch Training Run	Start the QLoRA fine-tuning job on Server 1. Monitor loss curves, learning rate schedule, and GPU utilization. Training runs for approximately 18-20 hours over 3 epochs.	ML Engineer
4.3	Checkpoint Saving	Save model checkpoints every 500 steps. If training diverges or quality drops, roll back to a previous checkpoint. All checkpoints logged to MLflow / W&B experiment tracker.	ML Engineer
4.4	Validation Monitoring	Monitor validation loss in real time. If validation loss stops decreasing for 3 consecutive checkpoints (early stopping), halt training to prevent overfitting.	ML Engineer
4.5	Security Penetration Test	Security team attempts to extract training data through adversarial prompts, probe model	IT Security

		endpoints for data leakage, verify VLAN isolation, and test access control enforcement.	
4.6	Access Control Validation	Test role-based access: auditor role cannot access raw training data, manager role cannot modify model weights, admin logs are verified complete. MFA enforcement confirmed.	IT / Compliance
4.7	Adapter File Export	Export the final fine-tuned adapter file (~50 MB). Store in versioned model registry (MLflow). Tag with training dataset version, date, and performance metrics.	ML Engineer

What is Experiment Tracking?	Every training run is logged — which data was used, what settings were chosen, how well it performed. Tools like MLflow or Weights & Biases create a complete history so we can always recreate any previous model version and compare runs side by side.
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PHASE 5	Evaluation & On-Premise Deployment	Weeks 9 – 10
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What Happens in This Phase

The fine-tuned model is evaluated by senior auditors against real workpaper examples it has never seen. If it passes quality thresholds, it is deployed on-premise via Ollama and opened to 5 pilot audit engagements. Auditors begin using it in real work, providing the first round of feedback.

Step	Action	Detail	Owner
5.1	Holdout Evaluation	Run the model on the 10% test set it has never seen. Calculate F1 score, ROUGE score, hallucination rate, and output format consistency. Compare against baseline from Step 4.1.	ML Engineer
5.2	Senior Auditor Blind Review	5 senior auditors independently review 50 model outputs each (250 total) without knowing which came from the AI vs a human. Score on accuracy, reasoning quality, and professional tone.	Senior Auditors
5.3	RAG Layer Integration	Connect the eCFR (Electronic Code of Federal Regulations) text database as a retrieval layer. Model now grounds citations in real regulatory	ML Engineer

		text, preventing hallucinated regulation references.	
5.4	Ollama Deployment	Load base model + adapter onto Server 2 via Ollama. Configure internal REST API endpoint. Test response time targets (under 2 seconds). Confirm zero external network calls.	DevOps / ML Engineer
5.5	Auditor UI Rollout	Deploy the auditor-facing interface. Auditors upload or select an Excel workpaper, view AI analysis alongside the workpaper, and see the feedback buttons for rating responses.	Product / Frontend
5.6	Pilot Engagements	Open to 5 live audit engagements. All AI outputs are marked 'AI-Assisted Draft' and require mandatory human review before sign-off. Collect initial feedback.	Audit Team

What is RAG?	RAG (Retrieval Augmented Generation) means the model looks up real source documents before answering. When asked about a regulation, it retrieves the actual regulation text first, then answers based on that. This prevents the model from inventing regulation numbers or misquoting thresholds.
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PHASE 6	Human Feedback Collection & DPO Alignment	Ongoing — Post Week 10
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What Happens in This Phase

This phase never ends — it runs continuously after deployment. Auditors use the product, provide feedback on responses, and those corrections accumulate into training data for quarterly model improvement cycles. Over time the model becomes calibrated to the firm's exact judgment standards.

What is DPO?	DPO (Direct Preference Optimization) is how we teach the model firm-specific judgment — not just what finance looks like, but how YOUR firm specifically decides, escalates, and documents. It uses pairs of responses: the one your auditor preferred vs the one the model originally gave. The model learns to behave like your senior auditors.
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Feedback Collection — How It Works in the Product

Feedback Type	What the Auditor Does
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Thumbs Up	Clicks approve — confirms the AI output was correct. Takes 1 second. This is the most common feedback type.
Thumbs Down	Clicks reject — flags the response as incorrect. The system logs the rejection with the original model output for later review.
Edit & Correct	Auditor edits specific fields (risk level, regulation cited, recommendation). This correction becomes a DPO training pair — the most valuable feedback type.
Free Text Note	Auditor writes a short note explaining why the model was wrong. Used to identify patterns in model errors and guide future training data creation.

Feedback Data Structure — What Gets Captured

Field Captured	Why It Is Stored
Workpaper input (anonymized)	The exact input that generated the response — needed to recreate the training pair
Model output	What the model said — becomes the 'rejected' side of the DPO pair if corrected
Corrected output	What the auditor said it should have been — becomes the 'chosen' side of the DPO pair
Reviewer tier	Whether correction came from senior auditor, qualified auditor, or junior staff — higher tier corrections get more weight in training
Model version	Which version of the model produced the output — enables version-over-version quality tracking
Timestamp + reviewer ID (anonymized)	For audit trail and access logging — required for compliance and security

DPO Training Cycles — The Improvement Flywheel

Cycle	Timeline	What Happens	Expected Improvement
Cycle 1	Month 3-4	300-500 corrections accumulated. First DPO run on risk classification and compliance thresholds.	Model stops under-rating risk on uncleared items
Cycle 2	Month 6	800-1,000 corrections. DPO on finding documentation style and firm-specific escalation thresholds.	Recommendations match how firm partners think
Cycle 3	Month 9-12	2,000-3,000 cumulative corrections. Full coverage across all workpaper types, edge cases, and new industries.	Model catches issues junior auditors miss

Security Architecture & Data Governance

Five Security Layers

Layer	Name	What It Does
L1	Network Isolation	Air-gapped VLAN. No external API calls permitted. Firewall blocks model endpoints from internet. TLS 1.3 for internal traffic only.
L2	Authentication	LDAP/Active Directory integration. Role-based access (auditor / manager / admin). Multi-factor authentication enforced. Session timeout 15 minutes.
L3	Data Encryption	AES-256 encryption at rest for all workpaper data. TLS in transit. Encryption keys managed by firm HSM (Hardware Security Module) — not by any cloud vendor.
L4	Audit Trail	Every model query logged with user ID and timestamp. All outputs reviewed before workpaper entry. Full SIEM integration for security monitoring.
L5	Data Residency	Model weights stored on-premise only. No training data sent to cloud. Inference server inside firm data center. Client data never leaves the building — ever.

AI Output Policy

All AI outputs are marked 'AI-Assisted Draft' Mandatory human review is required before any AI output is entered into a final workpaper. The AI cannot auto-populate final workpapers. Every output requires auditor sign-off. All outputs are stored with a query hash for full traceability and forensic recovery if needed.

Observability, Experiment Tracking & Artifact Versioning

Every aspect of the model's lifecycle — training runs, performance metrics, model versions, and deployment history — is tracked and versioned. This ensures full reproducibility and a complete audit trail of model changes.

Tool	Layer	What It Tracks
MLflow / W&B	Experiment tracking	Every training run — loss curves, accuracy, hyperparameters, dataset version used, adapter file output
DVC	Data versioning	Every version of the training dataset — which workpaper files, how many pairs, when created. Links dataset to model checkpoint.
HuggingFace Hub (private)	Model registry	All adapter versions stored with model cards. Git-backed versioning. Private repo — no public access.
Langfuse (self-hosted)	Inference tracing	Every auditor query traced — input, output, latency, token count, feedback rating. Self-hosted to keep query data on-premise.
Prometheus + Grafana	Infrastructure	GPU utilization, server health, request rates, response time percentiles. Alerts on degradation.

Project Timeline — 10 Weeks at a Glance

Weeks	Phase	Key Deliverables	Milestone
1 – 2	Data Collection & Governance	Anonymized workpaper dataset, legal sign-off, VLAN configured, Excel-to-JSON pipeline running	Data cleared
3 – 4	Dataset Construction	5,000+ JSONL training pairs, multi-task tagged, senior auditor reviewed, train/val/test split complete	Dataset ready
5 – 6	Infrastructure & LoRA Config	Both servers configured, software stack installed, LoRA parameters set, baseline benchmark recorded	Infra live
7 – 8	Training Run & Security QA	3-epoch training complete, adapter file exported, penetration test passed, access controls validated	Model trained
9 – 10	Evaluation & Deployment	Senior auditor eval complete, Ollama deployed on-premise, RAG layer live, 5 pilot engagements open	Live in firm

Ongoing	Feedback & DPO Cycles	Quarterly DPO re-training on accumulated auditor corrections, model improves continuously	Flywheel
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Risk Register

Risk	Level	Description	Mitigation
Data Privacy Breach	HIGH	Client data inadvertently included in training, exposing PII through model outputs	Automated PII scrubbing + manual sample review + air-gapped VLAN + annual pen test
Model Hallucination	MEDIUM	Model invents regulation numbers or cites incorrect thresholds in audit findings	RAG grounding layer on eCFR text + mandatory human review before workpaper entry
Auditor Adoption	MEDIUM	Auditors distrust AI output or bypass the tool, reducing feedback quality	Pilot with 5 engagements first, show quality improvement over time, involve senior auditors early
Scope Creep	MEDIUM	Requests to expand to new workpaper types before core system is stable	Hard phase gates — no new workpaper types until pilot engagements complete and DPO Cycle 1 done
Training Data Quality	LOW	Poor quality training pairs lead to poor model performance post fine-tuning	Senior auditor review of 500+ pairs before training, holdout eval set validates quality

Success Metrics

The project is considered successful when the following measurable outcomes are achieved:

Metric	Target	Measurement Method
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Risk classification accuracy	F1 $\geq$ 0.87	Holdout test set evaluation against senior auditor ground truth labels
Hallucination rate on regulation cites	$\leq$ 2%	100 adversarial queries testing invented regulation numbers, verified against eCFR source
Senior auditor approval rate (blind review)	$\geq$ 80%	250 outputs blind-reviewed by 5 senior auditors in Phase 5
Response latency	< 2 seconds	p95 response time measured under 50 concurrent queries on Server 2
Zero data exfiltration	0 incidents	Annual penetration test + continuous SIEM monitoring
Auditor feedback volume (Cycle 1)	$\geq$ 500 pairs	Feedback database count of correction-type feedback from Tier 1 and Tier 2 reviewers